

Q2. (Correcting the Pseudocode) [5 Marks]

The pseudocode below attempts to calculate the sum of digits of a number until a single digit is obtained (digital root). It contains FOUR logical/syntax errors. Identify and correct each error, stating what is wrong. text

```
BEGIN
INPUT num
SET result = num
WHILE result > = 10
SET sum = 0
WHILE result > 0
SET digit = result MOD 10
sum = sum + digit
result = result / 10
ENDWHILE
result = sum
ENDWHILE
OUTPUT "Digital root is", num
END
```

1. **Syntax Error (Line 4):** WHILE result > = 10
 - **What is wrong:** There is an invalid space between the greater-than (>) and equals (=) signs. The correct operator is >=.
2. **Syntax Error / Inconsistency (Lines 8, 9, 11):** sum = sum + digit, etc.
 - **What is wrong:** The pseudocode establishes a convention of using the SET keyword for variable assignment on lines 3, 5, and 7, but completely drops it on lines 8, 9, and 11.
3. **Logical Error (Line 9):** result = result / 10
 - **What is wrong:** Using a standard division symbol (/) will result in a floating-point decimal (e.g., \$15 / 10 = 1.5\$). To isolate digits, this needs to be **integer division** (dropping the remainder) so the loop eventually terminates. In pseudocode, this is typically written as DIV (or // in Python).

4. **Logical Error (Line 13):** OUTPUT "Digital root is", num

- **What is wrong:** The program calculates the final answer and stores it in the result variable, but the output statement accidentally prints num (the original, unmodified user input).

Corrected Pseudocode:

BEGIN

INPUT num

SET result = num

WHILE result >= 10

SET sum = 0

WHILE result > 0

SET digit = result MOD 10

SET sum = sum + digit

SET result = result DIV 10

ENDWHILE

SET result = sum

ENDWHILE

OUTPUT "Digital root is", result

END